

TRYHACKME LAB REPORT

(Networking Fundamentals & Intro to LAN)

Submitted By

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Course

Cybersecurity / Networking

Lab Platform

TryHackMe

Topics Covered

What is Networking

What is Internet

Identifying Devices on a Network

Ping (ICMP)

Introduction to LAN

Network Topologies

Subnetting Basics

ARP

DHCP

Date of Submission

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This report presents the practical learning and understanding of networking concepts gained through hands-on labs performed on the TryHackMe platform.

Introduction

In today's digital world, networking plays a very important role in how devices communicate and share information. Whether it is browsing websites, sending emails, or using online applications, everything depends on networks working properly. For anyone interested in cybersecurity or IT support, understanding networking basics is the first and most important step.

To build this foundation, I completed a series of hands-on labs on the TryHackMe platform. TryHackMe provides practical, beginner-friendly environments where concepts can be learned by doing instead of just reading. This made it easier for me to understand how networks actually work in real-life situations.

In this report, I have documented my learning from the networking-related rooms, mainly "*What is Networking*" and "*Intro to LAN*." These labs helped me understand how devices connect with each other, how data is transferred, and how different protocols like IP, ARP, and DHCP are used in communication.

While performing these labs, I also practiced basic commands like ping to check connectivity between systems. This gave me a clearer idea of how troubleshooting works in real networks.

Overall, these labs helped me strengthen my basic networking knowledge and gave me confidence to move forward in areas like cybersecurity, system administration, and IT support.

Room 1: What is Networking

Objective

The main aim of this room was to understand the basic idea of networking and how devices connect and communicate with each other.

Overview

Networking is simply connecting two or more devices so they can share information. These devices can be computers, mobile phones, servers, or any other smart device.

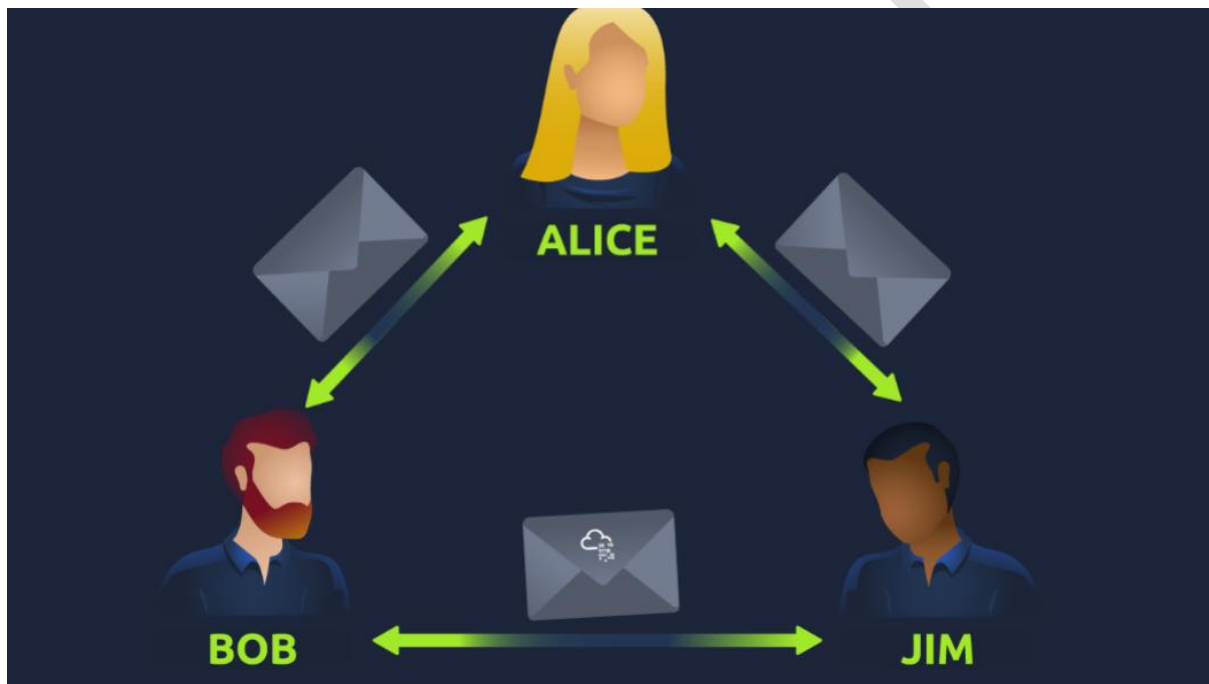
In simple words, whenever we send a message, open a website, or download a file, networking is working in the background.

This room helped me understand how devices talk to each other using IP addresses and how data travels from one system to another.

What I Learned

1. What is Networking

Networking means connecting devices so they can share data and resources. For example, in a college lab, all computers are connected so files and internet can be shared.

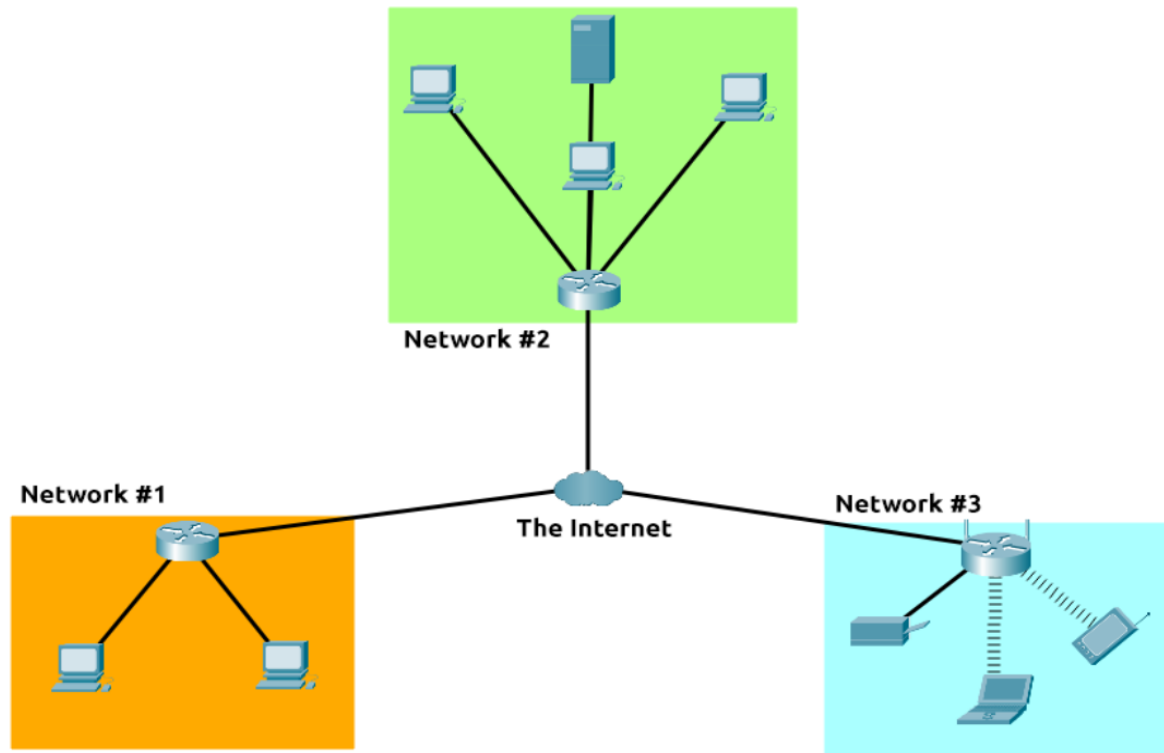


Q. What is the key term for devices that are connected together?

Ans - Network

2. What is the Internet

The internet is a big network made by connecting many smaller networks across the world. It helps us access websites, social media, emails, and many online services.



Q. Who invented the World Wide Web?

Ans – Tim Berners-Lee

He was a British computer scientist working at CERN (European Organization for Nuclear Research) in Switzerland.

3. Identifying Devices on a Network

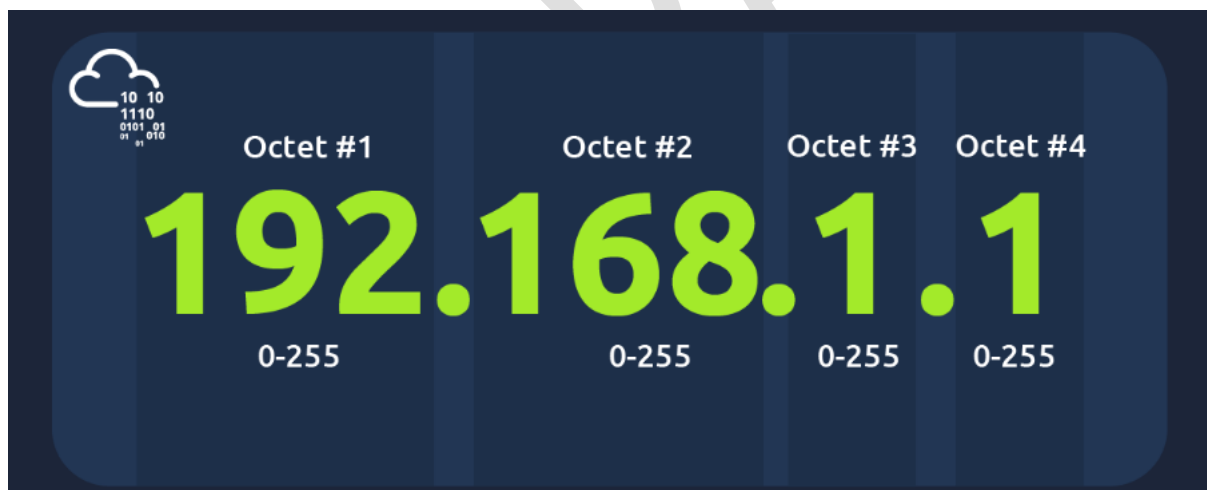
Every device connected to a network has its own identity:

- **IP Address** → used to identify a device in a network
- **MAC Address** → a unique hardware address of a device

These help in sending data to the correct system.

An **IP** address is a set of numbers that are divided into four octets. The value of each octet will summarise to be the IP address of the device on the network.

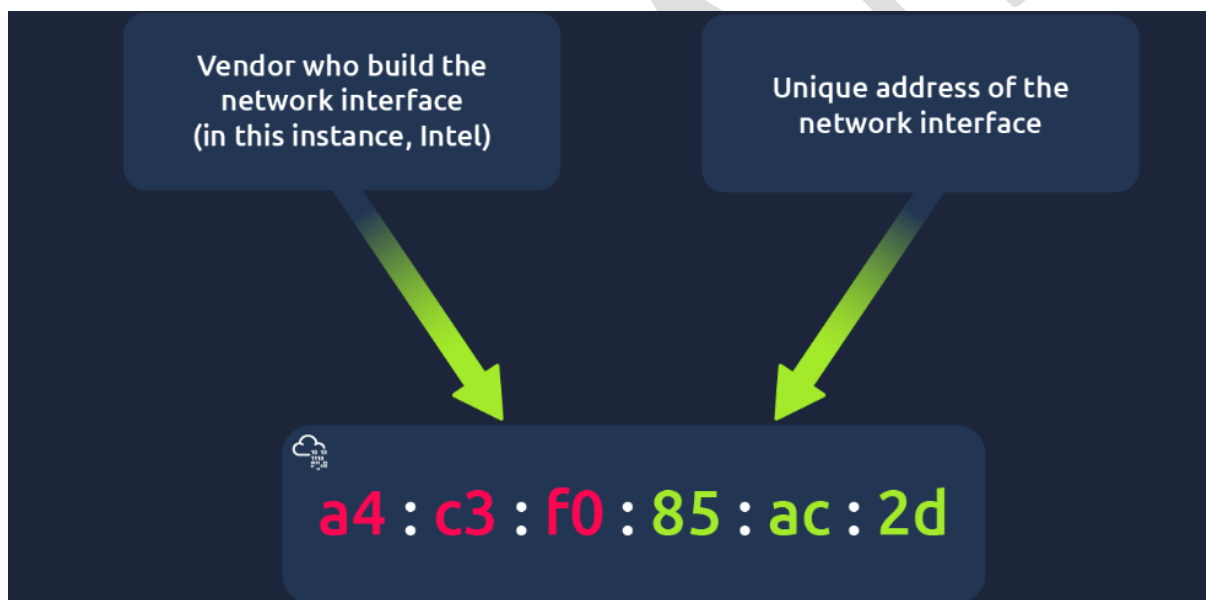
This number is calculated through a technique known as IP addressing & subnetting, but that is for another day. What's important to understand here is that IP addresses can change from device to device but cannot be active simultaneously more than once within the same network.



IP Addresses follow a set of standards known as protocols. These protocols are the backbone of networking and force many devices to communicate in the same language, which is something that we'll come onto another time. However, we should recall that devices can be on both a private and public network. Depending on where they are will determine what type of IP address they have: a public or private IP address.

MAC Addresses

Devices on a network will all have a physical network interface, which is a microchip board found on the device's motherboard. This network interface is assigned a unique address at the factory it was built at, called a **MAC (Media Access Control)** address. The MAC address is a **twelve-character** hexadecimal number (*a base sixteen numbering system used in computing to represent numbers*) split into two's and separated by a colon. These colons are considered separators. For example, *a4:c3:f0:85:ac:2d*. The first six characters represent the company that made the network interface, and the last six is a unique number.



Q. What does the term "IP" stand for?

Ans - Internet Protocol

Q. What is each section of an IP address called?

Ans - octet

Q. How many sections (in digits) does an IPv4 address have?

Ans - 4

Q. What does the term "MAC" stand for?

Ans – Media Access Control

Q. Deploy the interactive lab using the "View Site" button and spoof your MAC address to access the site. What is the flag?

Ans - THM{YOU_GOT_ON_TRYHACKME}

4. Ping Command

Ping is a simple tool used to check if a device is connected or not.

It sends a small message to another device and waits for a reply. If the reply comes back, it means the device is reachable.

Ping is one of the most fundamental network tools available to us. Ping uses **ICMP (Internet Control Message Protocol)** packets to determine the performance of a connection between devices

```
cmnatic@CMNatic-THM-LPTOP:~$ ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data.
64 bytes from 192.168.1.254: icmp_seq=1 ttl=63 time=2.18 ms
64 bytes from 192.168.1.254: icmp_seq=2 ttl=63 time=2.53 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=63 time=2.15 ms
64 bytes from 192.168.1.254: icmp_seq=4 ttl=63 time=3.34 ms
64 bytes from 192.168.1.254: icmp_seq=5 ttl=63 time=10.3 ms
64 bytes from 192.168.1.254: icmp_seq=6 ttl=63 time=4.45 ms
^C
--- 192.168.1.254 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 2.152/4.160/10.313/2.864 ms
cmnatic@CMNatic-THM-LPTOP:~$
```

Here we are pinging a device that has the private address of *192.168.1.254*. Ping informs us that we have sent six ICMP packets, all of which were received with an average time of 4.16 milliseconds.

Q. What protocol does ping use?

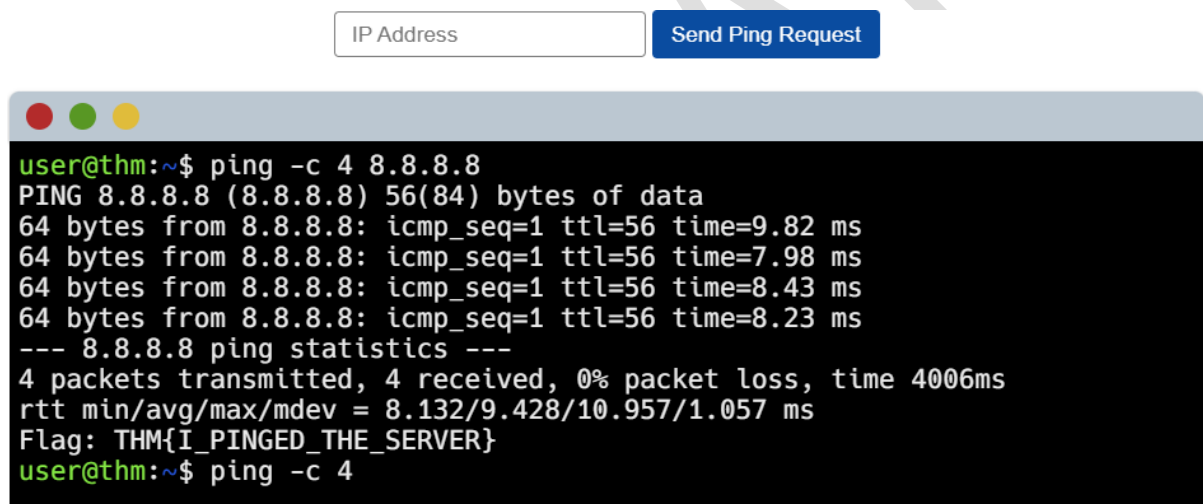
Ans - ICMP

Q. What is the syntax to ping 10.10.10.10?

Ans – ping 10.10.10.10

Q. What flag do you get when you ping 8.8.8.8?

Ans - THM{I_PINGED_THE_SERVER}



Conclusion

From this room, I understood the basic concept of networking and how important it is in everyday life. I also learned how devices communicate and how tools like ping help in checking network connectivity. This knowledge is useful for anyone starting in IT or cybersecurity.

Room 2: Introduction to LAN

Objective

The aim of this room was to understand how Local Area Networks (LAN) work and learn the basic concepts like topologies, ARP, DHCP, and subnetting.

Overview

A Local Area Network (LAN) is a network that connects devices within a small area like a home, office, or college lab. It allows devices to communicate with each other and share resources like internet, files, and printers.

This room helped me understand how devices are organized in a network and how data moves between them.

What I Learned

1. LAN (Local Area Network)

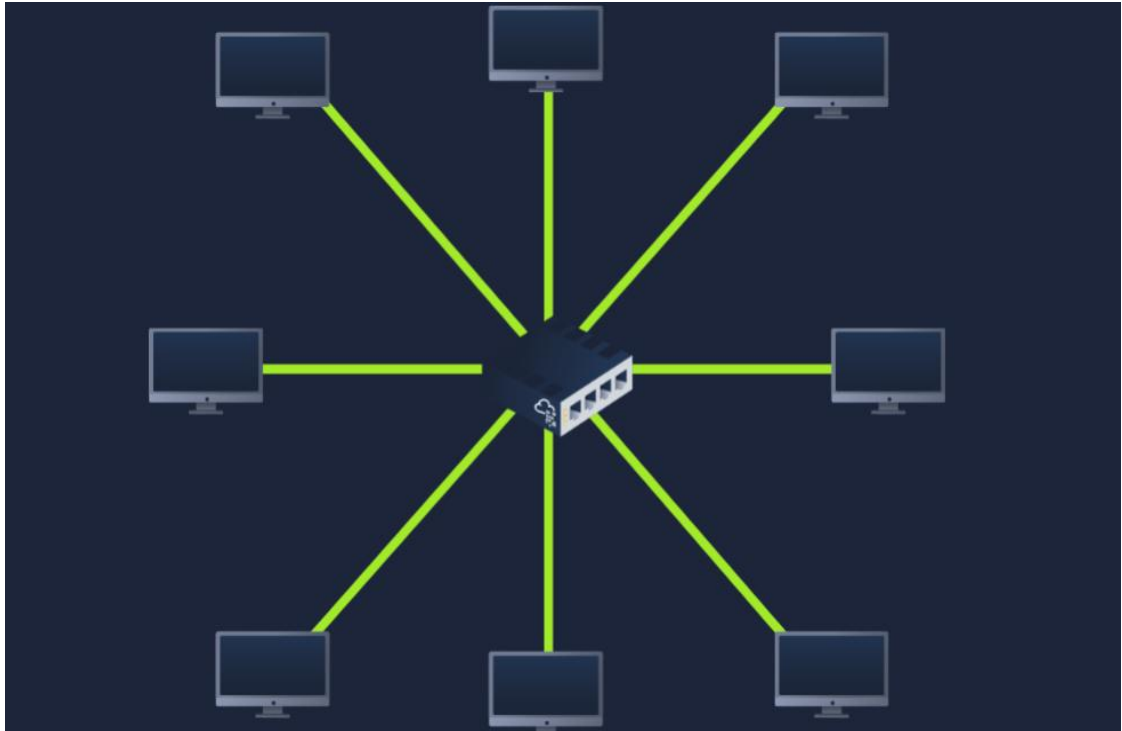
A LAN is a small network where devices are connected within a limited area. For example, all computers in a computer lab are part of a LAN.

LAN helps in fast communication and easy sharing of resources.

2. Network Topologies

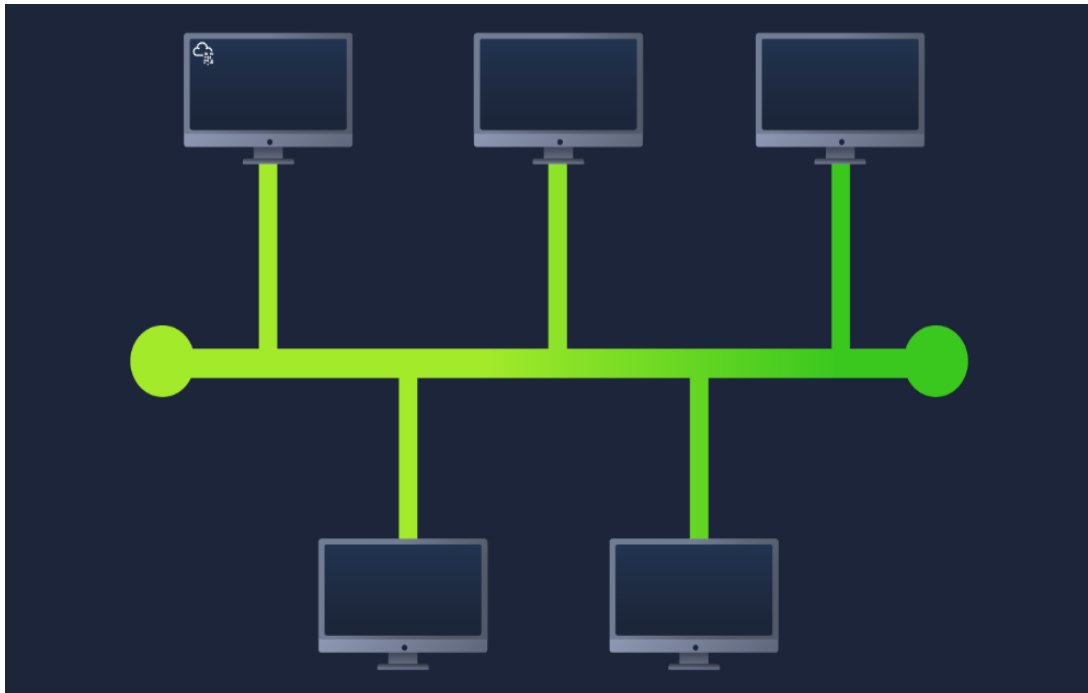
Topology means how devices are arranged in a network.

- **Star Topology** → All devices are connected to a central device (like a switch). This is the most commonly used setup.

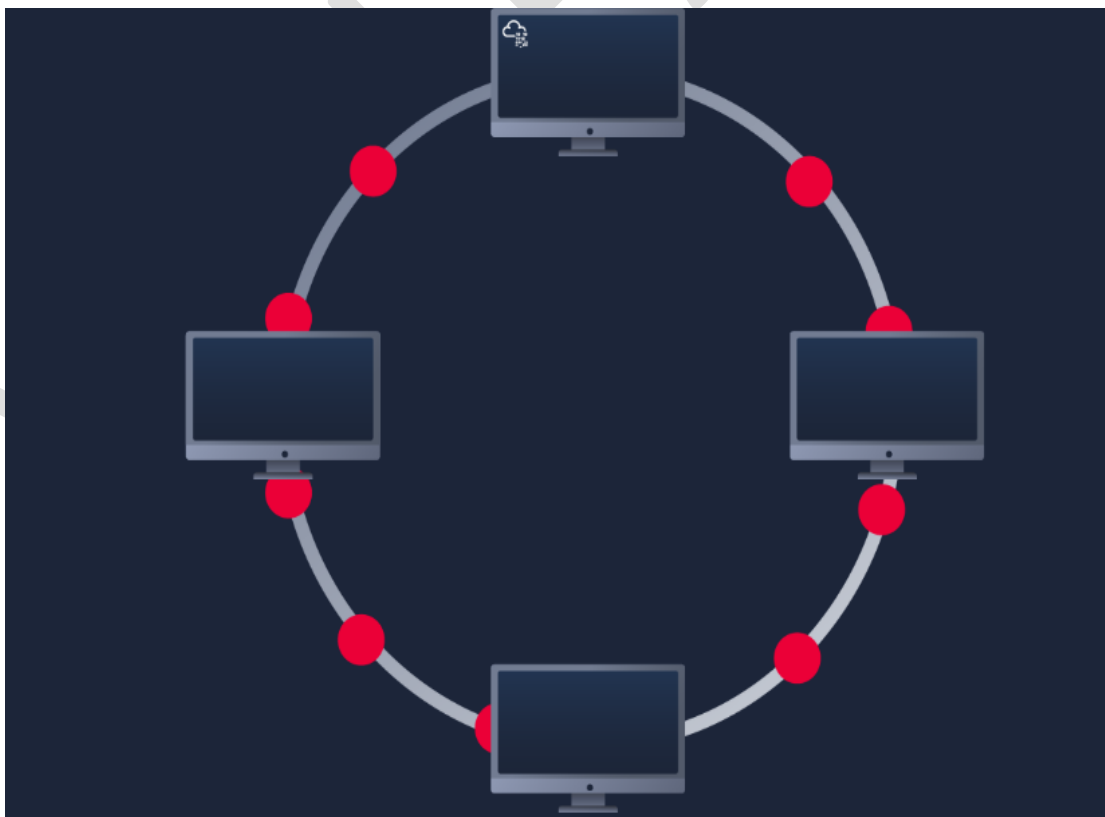


- **Bus Topology** → All devices are connected in a single line.

This type of connection relies upon a single connection which is known as a backbone cable. This type of topology is similar to the leaf off of a tree in the sense that devices (leaves) stem from where the branches are on this cable.



- **Ring Topology** → Devices are connected in a circular format.



Among these, star topology is widely used because it is easier to manage and troubleshoot.

Q. What does LAN stand for?

Ans – Local Area Network

What is the verb given to the job that Routers perform?

Ans - routing

Q. What device is used to centrally connect multiple devices on the local network and transmit data to the correct location?

Ans - Switch

Q. What topology is cost-efficient to set up?

Ans – Bus Topology

Q. What topology is expensive to set up and maintain?

Ans – Star Topology

Q. Complete the interactive lab attached to this task. What is the flag given at the end?

Ans- THM{TOPOLOGY_FLAWS}



3. ARP (Address Resolution Protocol)

ARP is used to find the MAC address of a device using its IP address.

In simple words, when one device wants to communicate with another, ARP helps in finding the physical address of that device inside the network.

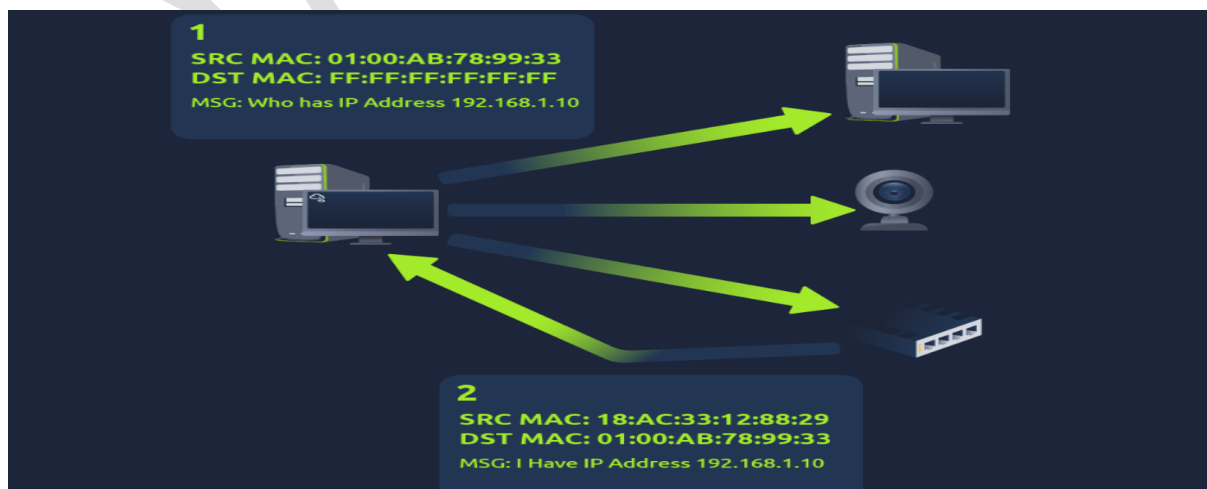
How does ARP Work?

Each device within a network has a ledger to store information on, which is called a cache. In the context of **ARP**, this cache stores the identifiers of other devices on the network.

In order to map these two identifiers together (IP address and MAC address), ARP sends two types of messages:

1. **ARP Request**
2. **ARP Reply**

When an **ARP request** is sent, a message is broadcasted on the network to other devices asking, "What is the mac address that owns this IP address?" When the other devices receive that message, they will only respond if they own that IP address and will send an **ARP reply** with its MAC address. The requesting device can now remember this mapping and store it in its **ARP cache** for future use.



Q. What does ARP stand for?

Ans – Address Resolution Protocol

Q. What category of ARP Packet asks a device whether or not it has a specific IP address?

Ans - Request

Q. What address is used as a physical identifier for a device on a network?

Ans – MAC Address

Q. What address is used as a logical identifier for a device on a network?

Ans – IP Address

4. DHCP (Dynamic Host Configuration Protocol)

DHCP is used to automatically assign IP addresses to devices in a network.

Instead of manually setting IP addresses, DHCP does this automatically, which makes network management easier.

DORA Process (How DHCP Works)

DHCP works in 4 simple steps called **DORA**:

1. Discover

When a device connects to a network, it first sends a message called **DHCP Discover**.

👉 Meaning:

“I am a new device, is there any DHCP server available?”

2. Offer

The DHCP server responds with a **DHCP Offer**.

👉 Meaning:

“I am available, and I can give you this IP address.”

3. Request

The device then sends a **DHCP Request** message.

👉 Meaning:

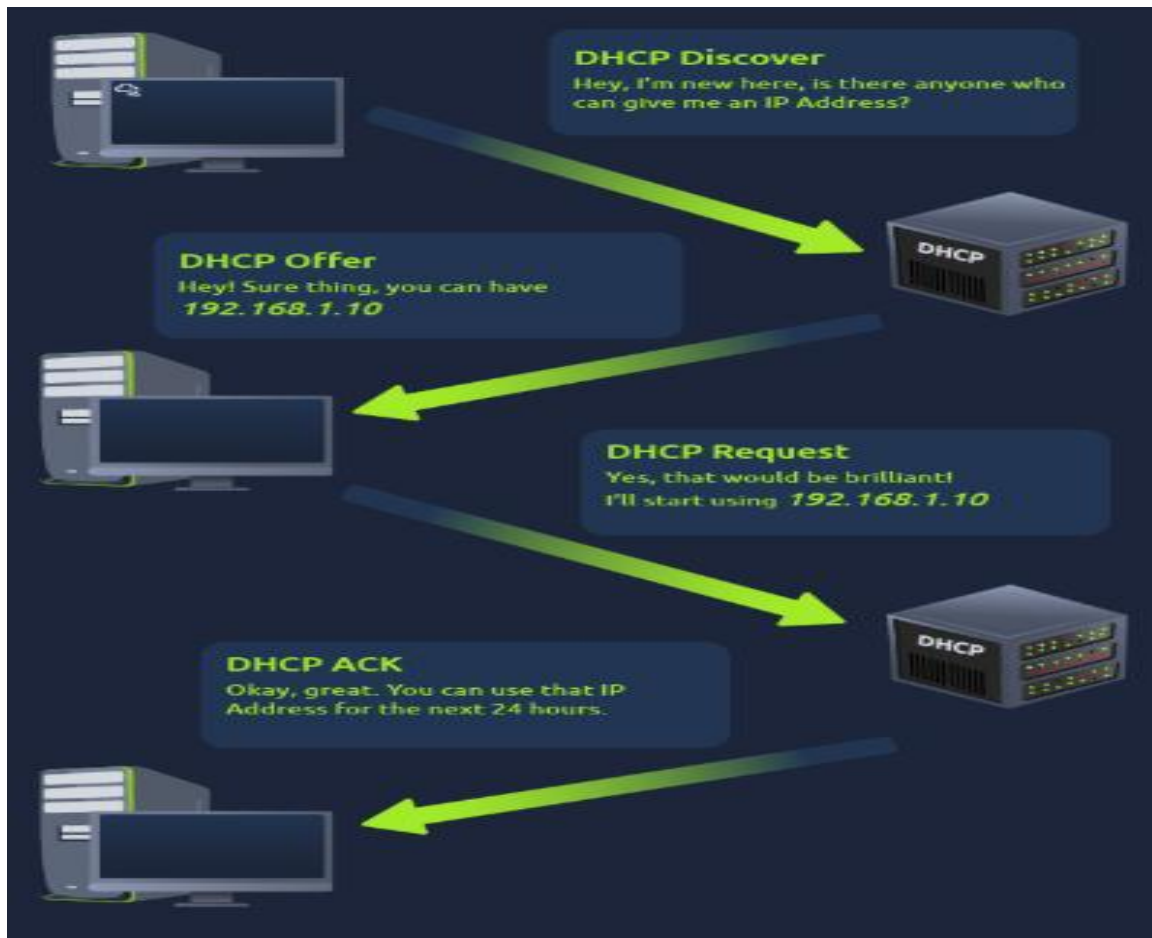
“I accept this IP address, please assign it to me.”

4. Acknowledge

Finally, the DHCP server sends a **DHCP Acknowledge (ACK)** message.

👉 Meaning:

“IP address is now assigned to you, you can use the network.”



Q. What type of DHCP packet is used by a device to **retrieve an IP address**?

Ans - DHCP Discover

Q. What type of DHCP packet does a device **send once it has been offered an IP address** by the DHCP server?

Ans - DHCP Request

Q. Finally, **what is the last** DHCP packet that is sent to a device from a DHCP server?

Ans -DHCP ACK

5. Subnetting (Basic Idea)

Subnetting means dividing a large network into smaller parts.

This helps in:

- Better performance
- Better security
- Easy management of devices

It is commonly used in large organizations.

Subnetting means dividing a big network into smaller parts.

It is done so that the network becomes easier to manage and works faster.

Simple Example

If there are 100 computers in one network, instead of keeping them all together, we divide them into smaller groups like:

- Group 1
- Group 2
- Group 3

Each group works like a separate small network.

Why it is used

- To manage network easily
- To reduce traffic
- To improve performance

Easy way to remember

👉 Big network → small networks = Subnetting

Q. What is the technical term for dividing a network up into smaller pieces?

Ans - subnetting

Q. How many bits are in a subnet mask?

Ans - 32

Q. What is the range of a section (octet) of a subnet mask?

Ans - 0-255

Q. What address is used to identify the start of a network?

Ans - Network Address

Q. What address is used to identify devices within a network?

Ans - Host Address

Q. What is the name used to identify the device responsible for sending data to another network?

Ans - Default Gateway

Conclusion

This room helped me understand how small networks like LAN work in real life. I learned how devices are arranged, how IP addresses are assigned automatically, and how devices find each other using ARP. These concepts are very important for networking and cybersecurity basics.

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